DATA ANALYTICS PROJECT 3

SQL DATA ANALYSIS

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Introduction

Data analytics plays an important role in extracting meaningful information from raw data. SQL (Structured Query Language) is widely used for managing and analyzing large datasets. In this project, SQL queries were used to analyze an e-commerce dataset and generate useful business insights.

The purpose of this project is to demonstrate SQL fundamentals such as data retrieval, filtering, sorting, grouping, and aggregation. Various SQL queries were executed to understand customer behavior, product performance, revenue generation, and order management.

Objective

The main objectives of this project are:

- To perform SQL-based data analysis.
- To retrieve data using SELECT statements.
- To filter data using WHERE clauses.
- To sort data using ORDER BY.
- To group data using GROUP BY.
- To perform aggregations using COUNT(), SUM(), and AVG().
- To generate business insights from transactional data.

Dataset Description

The dataset contains e-commerce transaction records. Each record represents an order placed by a customer.

The dataset includes the following attributes:

- OrderID
- Date
- CustomerID
- Product
- Quantity
- UnitPrice
- ShippingAddress
- PaymentMethod
- OrderStatus
- TrackingNumber
- ItemsInCart
- CouponCode
- ReferralSource
- TotalPrice

The dataset was imported into SQL Server Management Studio (SSMS) for analysis.

Tools Used

- Microsoft SQL Server Management Studio (SSMS)
- SQL
- Microsoft Excel

SQL Queries and Analysis

Query 1: Display All Records

```
SELECT * FROM orders;
```

Purpose:
This query displays all records from the orders table.

Query 2: Total Number of Orders

```
SELECT COUNT(*) AS TotalOrders

FROM orders;
```

Purpose:
This query calculates the total number of orders available in the dataset.

Query 3: Total Revenue

```
SELECT SUM(TotalPrice) AS TotalRevenue

FROM orders;
```

Purpose:
This query calculates the total revenue generated from all orders.

Query 4: Average Order Value

```
SELECT AVG(TotalPrice) AS AverageOrderValue

FROM orders;
```

Purpose:
This query calculates the average order value.

Query 5: Orders by Status

```
SELECT OrderStatus,

    COUNT(*) AS TotalOrders

FROM orders

GROUP BY OrderStatus;
```

Purpose:
This query shows the number of orders under each order status category.

Query 6: Revenue by Product

```
SELECT Product,

    SUM(TotalPrice) AS Revenue
```

```sql
FROM orders
GROUP BY Product
ORDER BY Revenue DESC;
```

Purpose:
This query identifies the products generating the highest revenue.

Query 7: Quantity Sold by Product

```sql
SELECT Product,
    SUM(Quantity) AS QuantitySold
FROM orders
GROUP BY Product
ORDER BY QuantitySold DESC;
```

Purpose:
This query determines the best-selling products based on quantity sold.

Query 8: Revenue by Payment Method

```sql
SELECT PaymentMethod,
    SUM(TotalPrice) AS Revenue
FROM orders
GROUP BY PaymentMethod
ORDER BY Revenue DESC;
```

Purpose:
This query analyzes revenue generated through different payment methods.

Query 9: Orders by Referral Source

```sql
SELECT ReferralSource,
    COUNT(*) AS OrdersCount
FROM orders
GROUP BY ReferralSource
ORDER BY OrdersCount DESC;
```

Purpose:
This query identifies which referral sources generate the most customer orders.

Query 10: Top 10 Highest Value Orders

```
SELECT TOP 10 *

FROM orders

ORDER BY TotalPrice DESC;
```

Purpose:
This query displays the highest-value orders in the dataset.

Key Findings

- SQL helped efficiently analyze large volumes of e-commerce data.
- Aggregation functions provided quick business insights.
- Product-level analysis highlighted top-performing products.
- Revenue analysis helped evaluate overall business performance.
- Payment method analysis revealed customer purchasing preferences.
- Referral source analysis identified important customer acquisition channels.
- Order status analysis provided insight into order fulfillment performance.

Conclusion

This project successfully demonstrated the use of SQL for data analysis. Various SQL techniques including SELECT, GROUP BY, ORDER BY, COUNT(), SUM(), and AVG() were applied to extract meaningful insights from an e-commerce dataset.

The analysis improved understanding of customer transactions, product performance, revenue generation, and business operations. This project enhanced practical SQL skills and provided valuable experience in transforming raw data into actionable business intelligence.